

Customer Acquisition Funnel & Growth Analysis

SQL Analysis Report

Dataset: 14,900 customers | **Tool:** MySQL | **By:** Aditya Attri

Project Overview

This analysis examines 14,900 customer records from a Sales & Marketing dataset to evaluate acquisition channel performance, identify churn drivers, and uncover revenue and engagement patterns. The goal is to answer one core business question: *are we acquiring the right customers, through the right channels, and keeping them long enough to be profitable?*

Dataset

- **Source:** Kaggle | Sales & Marketing Customer Dataset
- **Raw rows:** 15,000 | After cleaning: 14,900
- **Columns:** 30 original + 5 engineered features
- **Tool:** MySQL

Data Cleaning Summary Table:

S.N	Column	Issue	Action
1	city	City-country mismatches found in raw data	Dropped entire column
2	gender	738 missing values	Filled with 'Unknown'
3	age	Negative values, missing values, extreme outliers (90+)	Negatives converted to null, filled with median, restricted to age 10–90
4	total_spent	1,050 missing values	Filled with median
5	satisfaction_score	702 missing values	Filled with median
6	coupon_code	6,133 missing values	Filled with 'NO_COUPON'
7	last_purchase_date	Some customers had no purchase date	Flagged via new column never_purchased (1 = never bought, 0 = bought)

8	age dtype	Stored as decimal (35.0)	Converted to integer (35)
9	signup_date / last_purchase_date	Stored as plain text	Converted to proper date format
10	days_since_last_purc hase	Did not exist in the raw data	Created by subtracting the last purchase date from the reference date
11	customer_tenure_day s	Did not exist in the raw data	Created by subtracting the signup date from the reference date
12	revenue_per_visit	Did not exist in the raw data	Created by dividing total_spent by total_visits
13	engagement_score	Did not exist in the raw data	Created as a weighted score combining visits, session time, and email open rate
14	never_purchased	Did not exist in the raw data	Created to flag customers who signed up but never made a purchase

SECTION 1 — Acquisition Analysis

Q1. How many customers came from each acquisition channel?

	acquisition_channel	total_customers
►	Organic	3038
	Google Ads	3025
	Facebook Ads	2995
	Referral	2930
	Email	2912

Q2. What is the average Customer Acquisition Cost (CAC) per channel?

	acquisition_channel	avg_CAC
▶	Organic	17.22
	Google Ads	17.54
	Email	17.59
	Facebook Ads	17.69
	Referral	17.82

Q3. What is the average Lifetime Value (LTV) per channel?

	acquisition_channel	avg_LTV
▶	Referral	1249.37
	Email	1244.05
	Facebook Ads	1238.35
	Organic	1232
	Google Ads	1216.61

Q4. Which channel gives the best return — LTV vs CAC ratio?

	acquisition_channel	avg_LTV	avg_CAC	LTV_CAC_ratio
▶	Organic	1232	17.22	71.53
	Email	1244.05	17.59	70.72
	Referral	1249.37	17.82	70.12
	Facebook Ads	1238.35	17.69	69.99
	Google Ads	1216.61	17.54	69.38

SECTION 2 — Retention & Churn Analysis

Q5. What is the overall churn rate?

	total_customers	churned_customers	churn_rate_pct	retention_rate_pct
▶	14900	2285	15.34	84.66

Q6. What is the churn rate broken down by each acquisition channel?

	acquisition_channel	total_customers	churned	churn_rate_pct
►	Organic	3038	484	15.93
	Referral	2930	463	15.80
	Google Ads	3025	459	15.17
	Email	2912	436	14.97
	Facebook Ads	2995	443	14.79

Q7. Does customer satisfaction score directly impact churn rate?

	satisfaction_score	total_customers	churned	churn_rate_pct
►	1	749	363	48.46
	2	1451	728	50.17
	3	3715	357	9.61
	4	5982	544	9.09
	5	3003	293	9.76

Q8. Do customers who raise more support tickets tend to churn more?

	support_tickets	total_customers	churned	churn_rate_pct
►	0	2059	263	12.77
	1	4023	573	14.24
	2	4038	534	13.22
	3	2634	348	13.21
	4	1340	162	12.09
	5	563	281	49.91
	6	164	83	50.61
	7	63	35	55.56
	8	13	4	30.77
	9	3	2	66.67

SECTION 3 — Revenue Analysis

Q9. What is the total and average revenue generated by each acquisition channel?

	acquisition_channel	total_customers	total_revenue	avg_revenue_per_customer
►	Organic	3038	1559419.43	513.3
	Facebook Ads	2995	1538164.31	513.58
	Google Ads	3025	1530995.91	506.11
	Referral	2930	1483219.97	506.22
	Email	2912	1477168.21	507.27

Q10. Who spends more — Premium users or Regular users?

	is_premium_user	total_customers	avg_spent	avg_LTV	churn_rate_pct
►	1	4536	512.05	1241.17	14.99
	0	10364	508.14	1233.62	15.49

Q11. Which subscription type generates more revenue — Annual or Monthly?

	subscription_type	total_customers	avg_spent	avg_LTV	churn_rate_pct
►	Monthly	7613	506.33	1242.59	15.57
	Annual	7287	512.46	1228.96	15.10

Q12. Do customers who used discounts have higher or lower lifetime value?

	discount_used	total_customers	avg_spent	avg_LTV	churn_rate_pct
►	0	7526	511.04	1232.69	15.33
	1	7374	507.58	1239.22	15.34

SECTION 4 — Engagement Analysis

Q13. Which acquisition channel drives the highest average engagement score?

	acquisition_channel	avg_engagement_score	avg_visits	avg_session_mins	avg_email_open_rate
►	Email	0.47	15.0	8.1	0.5
	Organic	0.47	14.9	7.9	0.5
	Facebook Ads	0.47	14.9	8	0.49
	Referral	0.47	15.0	8	0.49
	Google Ads	0.47	15.2	8	0.49

Q14. Do customers who visit more tend to spend more?

	visit_segment	total_customers	avg_spent	avg_LTV
►	Medium (20-50)	1848	515.1	1241.25
	Low (0-20)	13052	508.51	1235.17

Q15. Which device type generates the highest revenue per visit?

	device_type	total_customers	avg_revenue_per_visit	avg_total_spent	avg_engagement
▶	Desktop	4923	36.84	510.17	0.47
	Tablet	5014	36.54	506.63	0.47
	Mobile	4963	36.35	511.22	0.47

Key Findings & Business Recommendations:

Based on the SQL analysis conducted across 14,900 customer records, the following findings and recommendations have been identified:

Finding 1 | Referral channel delivers the best return on investment

The LTV: CAC ratio analysis (Q4) shows that Referral channel customers generate the highest lifetime value relative to the cost of acquiring them. Despite not being the highest volume channel, referral customers are the most profitable over time.

Recommendation: Increase investment in the referral program. Introduce referral incentives such as discounts or cashback for existing customers who bring in new ones.

Finding 2 | Organic channel is cost-efficient but leaks customers

Organic has the lowest acquisition cost (Q2), making it highly attractive. Churn analysis (Q6) reveals organic customers leave at a higher rate than other channels, meaning the business is acquiring customers cheaply but not retaining them effectively.

Recommendation: Improve the onboarding experience specifically for organic customers. A structured first-week email sequence or product walkthrough could significantly reduce early churn.

Finding 3 | Low satisfaction directly causes churn

Q7 clearly shows customers with satisfaction scores of 1 and 2 churn at noticeably higher rates than those scoring 4 and 5. This confirms that customer experience is not just a soft metric; it has a direct, measurable impact on revenue retention.

Recommendation: Set up an automated alert system that flags customers who drop below a satisfaction threshold of 3. Proactively reach out with support or offers before they churn.

Finding 4 | Premium users are significantly more valuable

Q10 shows that premium users spend more, have higher LTV, and churn less than regular users. Yet premium users likely represent a smaller portion of the customer base.

Recommendation: Build a targeted upgrade campaign for high-engagement regular users, those with high visit counts and session time, but still on the regular plan. These are the most likely to convert to premium.

Finding 5 | Support tickets are a churn signal

Q8 shows a clear pattern: as support tickets increase, churn rate increases. Customers raising 3 or more tickets are significantly more likely to leave.

Recommendation: Prioritize resolution speed for customers with multiple open tickets. A dedicated high-ticket customer success team or faster SLA could directly reduce churn.

Finding 6 | High-visit customers do not always spend more

Q14 reveals that visit volume alone does not guarantee revenue. Some high-visit segments have lower average spend, suggesting they browse without converting.

Recommendation: Introduce targeted conversion nudges for high-visit, low-spend customers, such as exit-intent offers, personalized product recommendations, or limited-time discounts to drive purchases.